

8th International Workshop on Quantum Compilation

19–20 September 2026

Toronto, Canada

Saturday Schedule

09:00	<i>Registration, Welcome, Coffee</i>
10:00	Vadym Kliuchnikov "TBA"
11:00	<i>Coffee break</i>
11:45	Pablo Andres-Martinez, Henry Semenenko, Maximilian Ruesch, Douglas Wilson, Callum Macpherson, Alec Edgington and Seyon Sivarajah <i>Fault-Tolerant Compilation of Guppy Programs</i>
12:10	Elaine Wong, Narasinga Rao Miniskar, Seyong Lee, Vicente Leyton-Ortega, Yufei Ding, Jeffrey S Vetter and Travis S Humble <i>MLIR-Based Backend-Agnostic Compiler for Fault-Tolerant Quantum Computing</i>
12:35	Bradley Chase and Farrokh Labib <i>CliffT: A Compiler-Backed Exact Simulator for Near-Clifford Fault-Tolerant Circuits</i>
13:00	<i>Lunch</i>
14:45	Andrey Boris Khesin, Sarah Meng Li, Boldizsár Poór, Benjamin Rodatz, John van de Wetering and Richie Yeung <i>SpiderCat: Optimal Fault-Tolerant Cat State Preparation</i>
15:10	Boldizsár Poór, Andrey Boris Khesin, Sarah Meng Li, Benjamin Rodatz, John van de Wetering and Richie Yeung <i>CSSCat: Scalable Fault-Tolerant CSS State Preparation</i>
15:35	Sam Heavey and Athena Caesura <i>A scheduler for the Active Volume architecture</i>
16:00	Julien Gacon and Matthew Treinish <i>Qiskit Resource Estimation</i>
16:25	<i>Coffee break</i>
17:00	POSTER SESSION
18:00	<i>Head to Queens Cross Food Hall for workshop dinner</i>

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Sunday Schedule

09:00	Coffee, Chatting, Hanging out
10:00	Aws Albarghouthi <i>"How to build a wicked fast quantum circuit optimizer"</i>
11:00	Coffee break
11:45	Prateek P Kulkarni <i>PassProbe: Quantifying Compiler Pass Contributions in Quantum Transpilation</i>
12:10	Mark Koch and Louis Narmour <i>Polyhedral Analysis of Quantum Programs</i>
12:35	Guillermo Alonso-Linaje, Korbinian Kottmann, David Wierichs and Nathan Killoran <i>Parameter-optimal unitary synthesis: flag decompositions and brick wall circuits</i>
13:00	Lunch
14:45	Richie Yeung, Aleks Kissinger and Rob Cornish <i>Equivariant Reinforcement Learning for Clifford Quantum Circuit Synthesis</i>
15:10	Vivien Vandaele <i>Asymptotically Optimal Quantum Circuits for Comparators and Incrementers</i>
15:35	Marius Bothe, Christoph Sünderhauf, Michael Witham, Earl Campbell and Nick Blunt <i>More efficient Clifford+T synthesis for small-angle rotations and application to Trotterization</i>
16:00	Coffee break
16:45	Johannes Frank, Boris Bourdoncle, Peter-Jan Derks and Théo Dessertaine <i>Two Layers, No Swaps: Biplanar SPOQC Architecture Improves Runtime of Fermi-Hubbard Simulation</i>
17:10	Simon Martiel, Timothée Goubault de Brugière, Tristan Cam and Ali Javadi-Abhari <i>A Quantum Cache model for modular quantum computing architectures</i>
17:35	Guillermo Alonso, Korbinian Kottmann, David Wierichs, Utkarsh Azad, David Ren, Catalina Alborno Anzola and Josh Izaac <i>Beyond "Quantum 101": an advanced educational content ecosystem for quantum compilation R&D</i>
18:00	Workshop over! IWQC26 Business Meeting in the pub